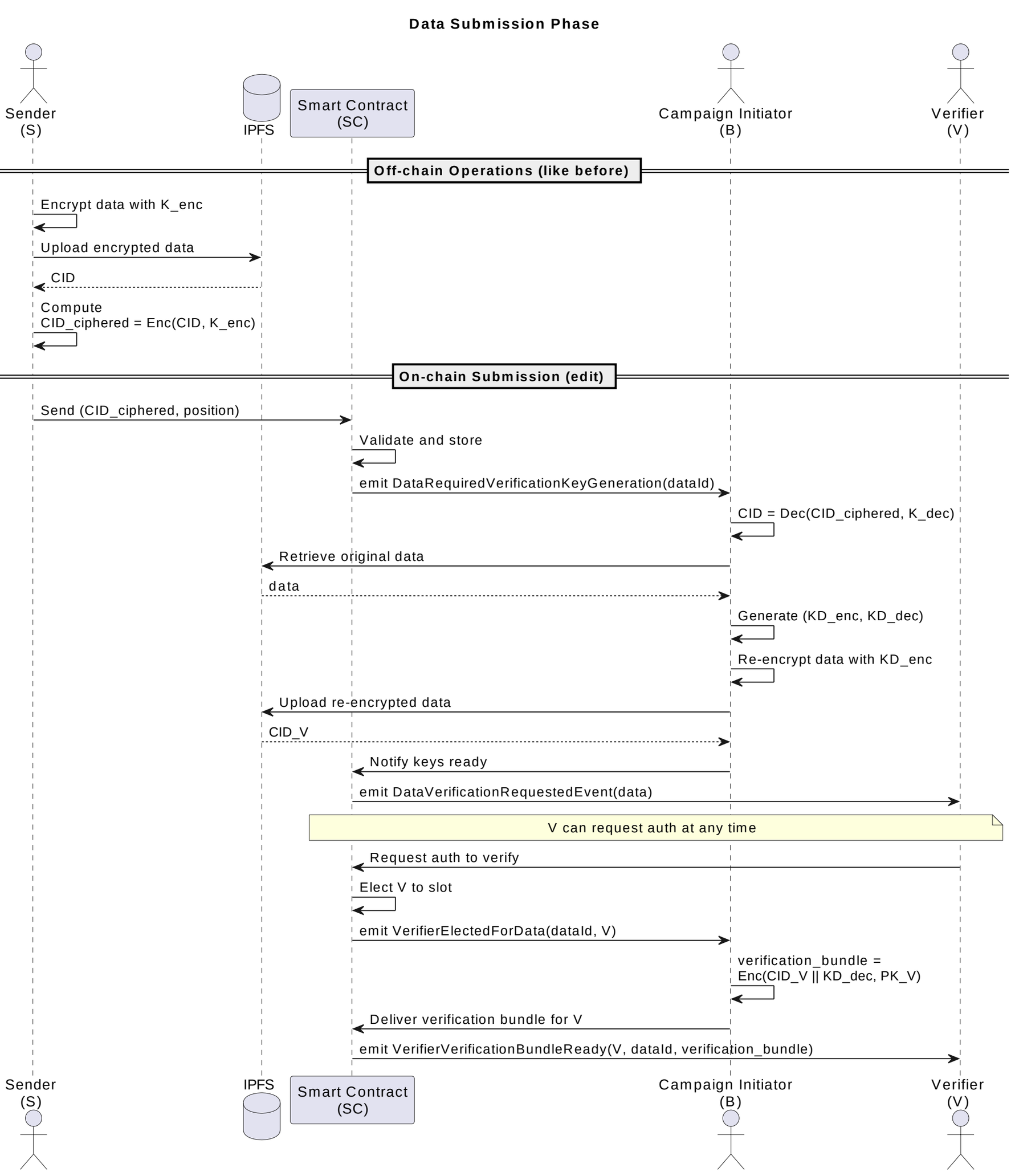
Data Submission Phase
Sender (S)
IPFS
Smart Contract (SC)
Campaign Initiator (B)
Verifier (V)
Off-chain Operations (like before)
Encrypt data with K_enc
Upload encrypted data
CID
Compute CID_ciphered = Enc(CID, K_enc)
On-chain Submission (edit)
Send (CID_ciphered, position)
Validate and store
emit DataRequiredVerificationKeyGeneration(dataId)
CID = Dec(CID_ciphered, K_dec)
Retrieve original data
data
Generate (KD_enc, KD_dec)
Re-encrypt data with KD_enc
Upload re-encrypted data
CID_V
Notify keys ready
emit DataVerificationRequestedEvent(data)
V can request auth at any time
Request auth to verify
Elect V to slot
emit VerifierElectedForData(dataId, V)
verification_bundle = Enc(CID_V || KD_dec, PK_V)
Deliver verification bundle for V
emit VerifierVerificationBundleReady(V, dataId, verification_bundle)
Sender (S)
IPFS
Smart Contract (SC)
Campaign Initiator (B)
Verifier (V)